1 · Stop the scam, not the customer

Northbridge Bank · Fraud Strategy & Operations · Proposal to the Executive Committee

- **£2.7m net annual value** (range £1.0m to £4.1m), mostly from APP-scam reimbursement cost Northbridge stops paying
- **94% of scam value intercepted** with the *same* number of payment holds (today: 58%)
- **13.7 → 2.2 genuine customers held per scam caught**: fewer good customers blocked
- **29,200 analyst hours released per year** through GenAI case notes and call scripts
- Working prototype on Databricks today; a 6-week shadow-mode pilot proves it on Northbridge's own data

> Synthetic data. Northbridge is a fictional UK retail bank (2.1M current-account customers). Model results are measured on a hold-out window. Bank-level £ figures apply those measured rates to published UK baselines. Every assumption is listed on slide 5.

2 · Why now: the PSR made APP scams a P&L line

- Since **7 Oct 2024**, UK banks **must reimburse** APP-scam victims, **up to £85,000 per claim**. The sending and receiving bank **split the cost 50:50**.
- UK APP-scam losses in 2024: **£450.7m** across **185,733 cases** (UK Finance, Annual Fraud Report 2025).
- PSR one-year update: **88% of losses on reimbursable claims returned** to victims (£112m in the first nine months).
- For Northbridge (≈2.6% of UK current accounts) that is **≈£11.7m of customer losses a year**, and roughly half of what's reimbursed now sits on our P&L.
- Today's control is a 3-condition rules engine. On our hold-out test it holds **1,533 payments to catch 58% of scam value**, with a hit-rate of **6.8%**. Most holds are genuine customers. Small purchase scams (half of all cases) go straight through.
- Consumer Duty raises the bar on *both* sides: protect vulnerable customers, and don't add needless friction for everyone else.

3 · What changes, for whom

	Executive sponsor: Chief Risk Officer / COO	Domain owner: Head of Fraud Operations
Today	Reimbursement liability growing, hard to forecast; regulator scrutiny on outcomes	A fixed team buried in false-positive holds; 11 minutes per case gathering evidence
With Scam Shield	**£1.7m/yr less net reimbursement cost**; one governed set of numbers for ExCo, the PSR and the board	**Same team, 94% of scam value caught**; every alert arrives with reason codes, a GenAI case note and a Consumer-Duty-safe call script
KPI moved	Net APP reimbursement cost (£), customer losses prevented (£), complaints	Value recall, precision (genuine holds per scam), minutes per case, SLA within 4 h
Proof	Business case on slide 5; measured on hold-out	Live queue in the app; decisions logged and fed back into the model

4 · Results on the hold-out test (10–29 Oct 2026, same alert budget)

Measure	Legacy rules	ML model	Change
Payments held	1,533	1,533	same budget
Scam value intercepted	58%	**94%**	**+36 pts**
Scams caught (count)	17%	**78%**	small purchase scams now caught
Hit-rate (precision)	6.8%	**31.4%**	4.6× better
Genuine customers held per scam	13.7	**2.2**	less friction
Holds needed to match today's catch	1,533	**168**	**-89%**

PR-AUC 0.609 · ROC-AUC 0.996 · LightGBM · value-weighted · time-based split · model v5 in Unity Catalog (`@champion`)

Scam type	Scam payments	Scam value	Rules value recall	Model value recall
investment	117	£1.4m	69%	98%
romance	106	£364,199	10%	99%
impersonation	74	£327,568	83%	90%
purchase	240	£136,381	9%	71%
invoice_redirection	24	£94,032	78%	75%
advance_fee	56	£19,596	0%	69%

5 · Business case (annual, Northbridge)

Line	Value	Basis
Scam value held by today's rules (hold-out test)	58.3%	Measured: model_evaluation, same alert budget
Scam value held by ML model (hold-out test)	94.5%	Measured: model_evaluation, same alert budget
Northbridge APP scam losses today (per year)	£11.7m	UK Finance 2024 APP losses GBP 450.7m x 2.6% PCA share (assumption)
Implied attempted scam value (per year)	£18.0m	losses / (1 - rules recall x 60% intervention success)
Customer losses prevented (per year)	£3.9m	attempted x (model recall - rules recall) x intervention success
Northbridge net reimbursement cost avoided	£1.7m	prevented x 88% reimbursed (PSR) x 50% after receiving-PSP share
Analyst hours released (per year)	29,200.0	1,200 holds/day x (11 -> 7 min) with GenAI case notes
Value of analyst capacity released	£1.1m	hours x GBP 38/h loaded cost (redeploy to complex cases)
Genuine customers held per scam caught: rules	13.7	Measured: 1/precision - 1
Genuine customers held per scam caught: model	2.2	Measured: Consumer Duty friction indicator
Platform run cost (per year)	£-180,000	Estimate: serverless pipeline + Lakebase + serving + FMAPI + app
Net annual value to Northbridge	£2.7m	net reimbursement avoided + analyst capacity - run cost

Sensitivity: model uplift held at its measured value; only the unvalidated assumptions move.

Scenario	UK PCA share	Intervention success	Minutes saved / case	Run cost	Customer losses prevented	Net annual value	ROI
Conservative	2.0%	40%	2	£300,000	£1.7m	**£1.0m**	4x
Base (headline)	2.6%	60%	4	£180,000	£3.9m	**£2.7m**	16x
Upside	3.0%	75%	5	£150,000	£6.5m	**£4.1m**	28x

ROI ≈ 16× run cost · payback ≈ 0.8 months. Assumptions marked as assumptions are validated in the pilot; measured figures come from the hold-out test.

6 · How it works: one governed journey, not five demos

- **Lakeflow**: Auto Loader + declarative pipeline, 7 data-quality expectations, a quarantine table, AUTO CDC de-duplication
- **Unity Catalog**: PII masked at source for everyone outside `northbridge-pii-readers` (including the AI); governed tags; automatic lineage; one metric view for every KPI
- **ML + GenAI**: a UC-registered model behind a real-time serving endpoint; Claude writes the case note, the call script and the customer warning
- **Lakebase**: millisecond lookups + transactional analyst decisions; decisions stream back to the lakehouse automatically
- **Genie**: the Head of Fraud asks questions in plain English over the same governed data
- **One bundle, one job**: reproducible, promotable to production

Proof it's one journey: payment `FP26001105668` (£8,552 to a 41-day-old EMI account) was *not* flagged by the legacy rules. The model scored it **CRITICAL 0.999**, Claude wrote the case note, an analyst stopped it in the app, and the decision flowed back into Unity Catalog seconds later. Every hop is shown with a query in `evidence/trace_one_payment.md`.

7 · What you'll see in the demo (tell → show → tell)

1. **Outcome**: model vs rules, £ business case, live ops counters
2. **Analyst queue**: a £-thousands "safe account" payment the rules **missed**; reason codes; GenAI case note and call script; one click records the decision (Lakebase)
3. **Live payment check**: the same customer pays a mule account. Lakebase lookups → model → "Pause and check" warning written by GenAI, in about a second
4. **Ask Genie**: "Which scam type cost us most in net reimbursement?" and a follow-up, with the SQL shown

8 · Governance, model risk and Consumer Duty

- **PRA SS1/23 model risk**: model inventory in UC, versioned with `@champion`, hold-out evaluation stored as a table, lineage from raw file to score, inference tables log every real-time decision
- **Explainability**: top-3 reason codes on every score, in plain English for analysts and customers
- **Human in the loop**: the model *holds and explains*; an analyst decides. Every decision is audited in Lakebase and in UC history
- **Data minimisation**: the operational store only gets masked names; Genie has PII columns excluded *and* masked
- **GenAI guardrails**: low temperature, strict JSON schema, non-accusatory open questions, failures logged and never blocking

9 · Decisions & trade-offs (summary)

- **Same-alert-budget comparison**, not AUC: it answers "more £ caught with the same team?"
- **Lakebase for the app, the warehouse for analytics**: point reads and transactions vs scans
- **Batch GenAI (Haiku) + interactive GenAI (Sonnet)**: cost where it's cheap, quality where it's customer-facing
- **Feature parity by code today; online feature serving in production**
- **Synthetic, scam-enriched data**: so we apply *rates*, not totals, to the business case. Re-measured in the pilot.

Full list: `docs/decisions-and-tradeoffs.md`

10 · Pilot: 6 weeks, shadow mode, no customer impact

Week	Activity	Exit criterion
1–2	Land 12 months of Northbridge payments + claims via Lakeflow; UC masks & grants with Security	Data reconciled with finance (claims £) within 1%
3–4	Retrain on Northbridge history; shadow-score live payments (no holds)	Value recall ≥ rules + 15 pts at equal holds
5	Analysts work the shadow queue in the app; GenAI notes evaluated by QA	≥ 25% handling-time reduction; zero non-compliant scripts
6	Model risk validation pack (SS1/23); ExCo go/no-go	Signed-off business case

What we need from Northbridge: a data owner from Fraud Ops, read access to the payments hub and complaints feeds, 2 analysts for week 5, and a model risk reviewer.

11 · Likely objections, and answers

Objection	Answer
"These are synthetic numbers."	Yes. That's why the business case applies *measured rates* to public baselines, and why the pilot re-measures on your data before go-live.
"Won't more holds annoy customers?"	Same number of holds, and 13.7 → 2.2 genuine customers per scam caught. Friction goes *down*.
"Can we explain a decline to the FOS?"	Every score carries reason codes; every analyst decision is time-stamped with the analyst and notes.
"Is GenAI safe in front of customers?"	It drafts; humans approve. Prompts enforce non-accusatory language. Outputs are logged and evaluated.
"Another database to run?"	Lakebase is serverless Postgres inside the same governance; sync both ways is managed, so there's no ETL to build.
"Lock-in?"	Delta/Parquet, open Postgres, MLflow, standard SQL.

12 · The ask

- **Approve a 6-week shadow-mode pilot** (Fraud Ops + Data + Model Risk)
- **Success = +15 pts value recall at equal holds and −25% handling time**, measured on Northbridge data
- **Decision at week 6**: production rollout plan, with real-time scoring in the payments journey